Marco Schouten

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Education

KTH Royal Institute of Technology — Stockholm, Sweden
Master of Science, Machine Learning, Grade 4.84/5.0 — 08/2020 - 10/2022

Delft University of Technology — Delft, Netherlands
Exchange, Computer Science — 08/2021 - 01/2022

University of Genoa — Genoa, Italy
Bachelor of Science, Computer Engineering, Grade 109/110 — 09/2017 - 07/2020

Experience

Aptiv PLC — Wuppertal, Germany
Research Assistant — 03/2023 - Present

- Developed scalable a machine learning algorithm for HD map generation from LiDAR data.
- Trained segmentation-based deep neural networks for accurate lane-level centerline extraction.
- Conducted literature reviews and collaborated across engineering teams for system deployment.

SMaRC Swedish Maritime Robotics Centre — Stockholm, Sweden
Research Assistant — 01/2022 - 09/2022

- Improved sonar-based pose estimation for underwater vehicles, achieving 1m accuracy.
- Analyzed sensor parameters and conducted experiments for system optimization.
- Validated proposed algorithm through extensive experiments and simulations.

SurgiQ Ltd — Genoa, Italy
Research Assistant — 03/2020 - 07/2020

- Developed an inductive AI-based hospital scheduling system, enhancing efficiency by 97%.
- Conducted research on inductive AI techniques, resulting in publications in AI*IA and CLIC.

Skills

Programming Languages	Python, Git
Machine Learning Frameworks	TensorFlow, PyTorch
Data Analysis Tools	Pandas, MySQL, MongoDB, Matplotlib
Languages	English (Fluent), Italian (Fluent), Dutch (basic), German (basic)

Awards

2022	**Selected at Nova**, a top 3% merit-based talent network for career development	Sweden
2020	**Hattrick Award in Artificial Intelligence**, awarded to 20% of the students [Link Award]	Sweden
2020	**Selected at Lead the Future**, mentorship with a 20% acceptance rate nationwide	Italy
2020	**Scholarship at University of Genoa 2018, 2019,2020**, ranked 4th out of 150+ students	Italy

Publications

Cademartori, L., Galatà, G., Monaco, C.L., Maratea, M., Mochi, M. and Schouten, M., **2022. An ASP-based Approach to Master Surgical Scheduling.** CILC 2022: 37th Italian Conference on Computational Logic

Alviano, M., Bertolucci, R., Cardellini, M., Dodaro, C., Galatà, G., Khan, M., Maratea, M., Mochi, M., Morozan, V., Porro, I. and Schouten, M., **2020. Answer Set Programming in Healthcare: Extended Overview.** IPS-RCRA@ AI*IA.

Projects

Real-Time Object Detection | KTH Stockholm, Sweden
Python 05/2022 - 06/2022

- Achieved real-time object detection at 45 frames per second (FPS) with a mean average precision (mAP) of 61%.
- Trained and optimized deep neural networks on a dataset of 10,000 images using YOLOv5s.
- [GitHub]

Human-Robot Skill Transfer | TU Delft Delft, Netherlands
Python 09/2021 - 12/2021

- Engineered ML-based software achieving 91% accuracy in transferring hand movements to a simulated manipulator.
- [GitHub]

Quadrotor Motion Planning | TU Delft Delft, Netherlands
Python 12/2021 - 01/2022

- Developed PRM path planner algorithm in Python for autonomous control, improving navigation accuracy.
- [GitHub]

CNN Training Time Estimation | TU Delft Delft, Netherlands
Python 08/2021 - 11/2021

- Optimized CNN training jobs, completing 33% more within a pre-defined timeline using cloud resources.
- [GitHub]